

# STT 861: Theory of Probability & Statistics I

## Lecture 14: Exam I Review & In-Class Exam

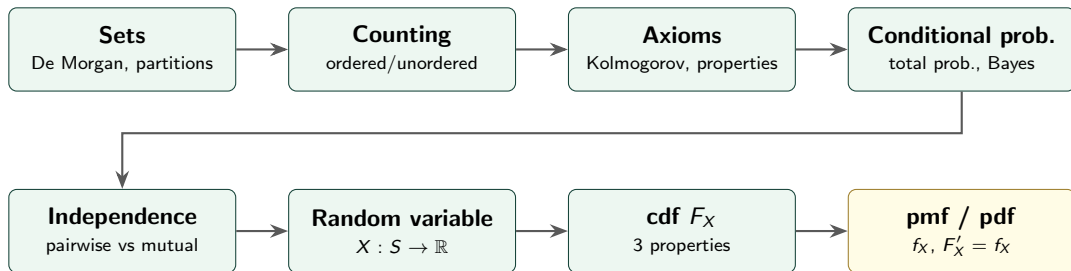
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Reference: Casella & Berger Ch. 1–2

# Chapter 1 map: from sets to random variables



Each box feeds the next: events are sets, probabilities of events obey the axioms, conditioning and independence describe how events interact, and random variables turn outcomes into numbers whose distribution is summarized by the cdf and pmf/pdf.

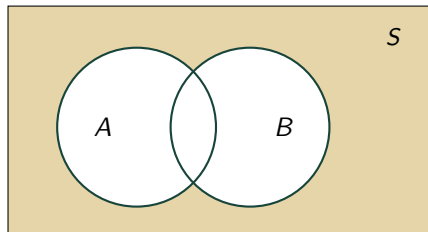
# Sets: operations, De Morgan, partitions

## Key facts (§1.1)

- Distributive:  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De Morgan:  $(A \cup B)^c = A^c \cap B^c$ ,  
 $(A \cap B)^c = A^c \cup B^c$
- $A, B$  disjoint if  $A \cap B = \emptyset$
- $A_1, A_2, \dots$  form a partition of  $S$  if pairwise disjoint and  $\bigcup_i A_i = S$

## Check: partition?

$S = \{1, \dots, 10\}$ ,  $A = \{1, 3, 5, 7, 9\}$ ,  $B = \{2, 3, 6, 9\}$ . Do  $A, A^c \cap B, A^c \cap B^c$  partition  $S$ ? Yes: they are disjoint and their union is  $S$ .



Shaded region =  $(A \cup B)^c = A^c \cap B^c$ .

## Counting: four cases

|           | Without replacement                  | With replacement   |
|-----------|--------------------------------------|--------------------|
| Ordered   | $\frac{n!}{(n-r)!}$                  | $n^r$              |
| Unordered | $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ | $\binom{n+r-1}{r}$ |

### Example: committee

5 men and 4 women; choose a committee of 3.

$$P(\text{exactly 2 women}) = \frac{\binom{4}{2}\binom{5}{1}}{\binom{9}{3}} = \frac{30}{84} = \frac{5}{14}.$$

### Example: with replacement, unordered

Ways to choose 3 scoops from 5 flavors (repeats allowed):  $\binom{5+3-1}{3} = \binom{7}{3} = 35$ .

**Ask first:** does order matter, and can an item be reused? That picks the cell of the table.

# Probability: axioms and consequences

## Kolmogorov axioms

- (1)  $P(A) \geq 0$  for every event  $A$ .    (2)  $P(S) = 1$ .    (3) For pairwise disjoint  $A_1, A_2, \dots$ :  $P(\bigcup_i A_i) = \sum_i P(A_i)$ .

## Consequences

- $P(\emptyset) = 0$ ,  $0 \leq P(A) \leq 1$
- $P(A^c) = 1 - P(A)$
- $A \subset B \Rightarrow P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(B) = \sum_i P(B \cap A_i)$  for a partition  $\{A_i\}$

## Inequalities

- **Boole**:  $P(\bigcup_i A_i) \leq \sum_i P(A_i)$
- **Bonferroni**:  $P(A \cap B) \geq P(A) + P(B) - 1$

## Example

$P(A) = .7$ ,  $P(B) = .6$ : then  $P(A \cap B) \geq .3$ .

# Conditional probability, total probability, Bayes' rule

## Definitions and rules

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0), \quad P(A \cap B) = P(A|B)P(B).$$

**Total probability** for a partition  $A_1, \dots, A_k$ :  $P(B) = \sum_i P(B|A_i)P(A_i)$ .

## Bayes' rule (CB Thm 1.3.5)

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_i P(B|A_i)P(A_i)}.$$

## Memory aid

Condition *shrinks the sample space to B*.  $P(A|B) \neq P(B|A)$  in general: they have the same numerator but different denominators.

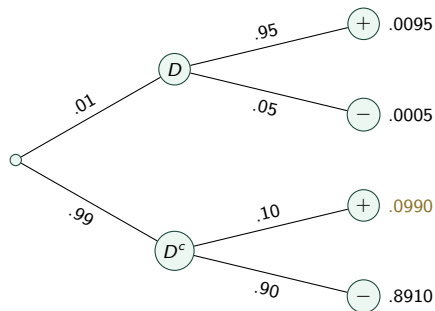
# Worked example: Bayes with a tree

A disease has prevalence 1%. A test is positive for 95% of the sick and for 10% of the healthy. Given a positive test, what is  $P(D|+)$ ?

$$P(D|+) = \frac{.01(.95)}{.01(.95) + .99(.10)} = \frac{.0095}{.1085} \approx .088.$$

## Moral

Even with a “good” test, most positives are false when the disease is rare.



Leaf values are joint probabilities;  
 $P(+) = .0095 + .0990 = .1085$ .

# Independence: pairwise versus mutual

## Definitions

$A, B$  **independent** if  $P(A \cap B) = P(A)P(B)$  (equivalently  $P(A|B) = P(A)$ ).

**Mutually independent**: the product rule holds for every sub-collection, e.g. also  $P(A \cap B \cap C) = P(A)P(B)P(C)$ .

If  $A \perp B$  then  $A \perp B^c$ ,  $A^c \perp B$ ,  $A^c \perp B^c$ .

## Pairwise $\nRightarrow$ mutual

Toss a fair coin twice.  $A = \text{H on toss 1}$ ,  $B = \text{H on toss 2}$ ,  $C = \text{both tosses equal}$ .

$P(A) = P(B) = P(C) = \frac{1}{2}$  and each pair has probability  $\frac{1}{4}$ , so pairwise independent. But  $P(A \cap B \cap C) = P(HH) = \frac{1}{4} \neq \frac{1}{8}$ .

## Do not confuse

**Disjoint** events with positive probabilities are *never* independent:  $P(A \cap B) = 0 \neq P(A)P(B)$ .

# Random variables and the cdf

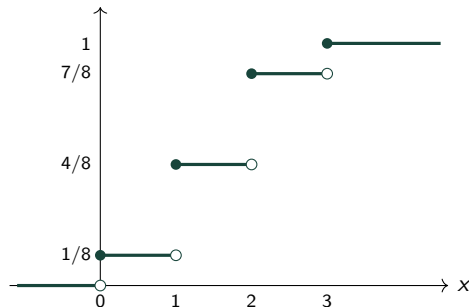
## Definitions (§1.4–1.5)

A **random variable** is a function  $X : S \rightarrow \mathbb{R}$ . Its **cdf** is  $F_X(x) = P(X \leq x)$ .

$F$  is a cdf iff

- (i)  $\lim_{x \rightarrow -\infty} F(x) = 0$  and  $\lim_{x \rightarrow \infty} F(x) = 1$
- (ii)  $F$  is nondecreasing
- (iii)  $F$  is **right-continuous**

Then  $P(a < X \leq b) = F(b) - F(a)$ .



cdf of  $X \sim \text{Bin}(3, \frac{1}{2})$ : jumps at 0, 1, 2, 3; filled dots show right-continuity;  $P(X = x) = F(x) - F(x^-)$ .

# pmf and pdf: properties and a worked example

## pmf (discrete) / pdf (continuous)

**pmf:**  $f_X(x) = P(X = x) \geq 0$ ,  $\sum_x f_X(x) = 1$ .

**pdf:**  $f_X \geq 0$ ,  $\int f_X = 1$ ,

$F_X(x) = \int_{-\infty}^x f_X(t)dt$ ,  $F'_X(x) = f_X(x)$  where  $F'$  exists.

$P(X = x) = 0$  for continuous  $X$ .

## Example

$f(x) = cx^2$  for  $0 < x < 2$ , 0 else.

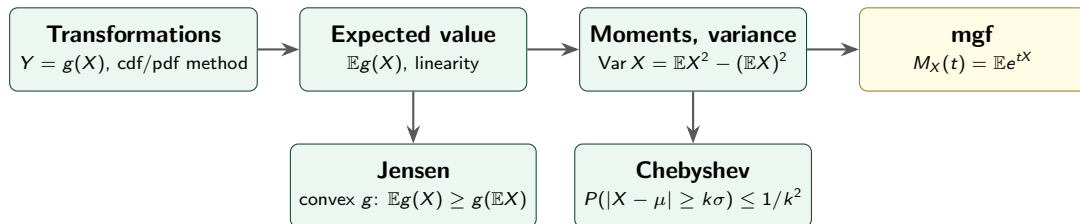
$$\int_0^2 cx^2 dx = \frac{8c}{3} = 1 \Rightarrow c = \frac{3}{8}.$$

$F(x) = \frac{x^3}{8}$  on  $(0, 2)$  (0 below, 1 above).

$$P(X > 1) = 1 - F(1) = \frac{7}{8}.$$

**Exam habit:** state the **support** explicitly and check that the density integrates to 1.

## Chapter 2 map: transformations, expectation, moments, mgf



Expectation turns a distribution into numbers; the mgf packages *all* moments in one function and identifies the distribution.

# Transformations $Y = g(X)$ : three methods

## cdf method (Thm 2.1.3)

$g$  increasing:  $F_Y(y) = F_X(g^{-1}(y))$ .

$g$  decreasing,  $X$  continuous:  $F_Y(y) = 1 - F_X(g^{-1}(y))$ .

Otherwise write  $F_Y(y) = P(g(X) \leq y)$  and solve for  $X$ .

## pdf method (Thm 2.1.5)

$g$  monotone on the support  $\mathcal{X}$ :

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|, \quad y \in \mathcal{Y}.$$

## Piecewise monotone (Thm 2.1.8)

If  $g$  is monotone on pieces  $A_1, \dots, A_k$  with inverses  $g_i^{-1}$ :

$$f_Y(y) = \sum_i f_X(g_i^{-1}(y)) \left| \frac{d}{dy} g_i^{-1}(y) \right|.$$

## Probability integral transform (Thm 2.1.10)

$X$  with *continuous* cdf  $F_X$ :  $U = F_X(X) \sim \text{Uniform}(0, 1)$ .

**Checklist:** (1) find the new support  $\mathcal{Y}$ , (2) invert  $g$ , (3) differentiate and take the **absolute value**, (4) substitute.

# Transformations: two worked examples

(1)  $X \sim U(0, 1)$ ,  $Y = -\ln X$

$g$  is decreasing;  $g^{-1}(y) = e^{-y}$ ;  $\mathcal{Y} = (0, \infty)$ .

$f_Y(y) = 1 \cdot |-e^{-y}| = e^{-y}$ ,  $y > 0$ .

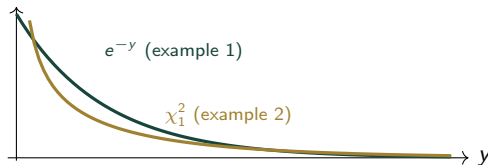
So  $Y \sim \text{Exponential}(1)$ : the inverse-transform recipe from Thm 2.1.10.

(2)  $X \sim N(0, 1)$ ,  $Y = X^2$

Not monotone: pieces  $x = \pm\sqrt{y}$ ,  $|\frac{dx}{dy}| = \frac{1}{2\sqrt{y}}$ .

$f_Y(y) = 2 \cdot \frac{e^{-y/2}}{\sqrt{2\pi}} \cdot \frac{1}{2\sqrt{y}} = \frac{e^{-y/2}}{\sqrt{2\pi y}}$ ,  $y > 0$ .

This is the  $\chi_1^2$  density.



# Expected value and Jensen's inequality

## Definition (§2.2)

$\mathbb{E}g(X) = \sum_x g(x)f_X(x)$  (discrete) or  $\int g(x)f_X(x)dx$  (continuous), provided  $\mathbb{E}|g(X)| < \infty$ .

**Linearity (CB Eq. 2.2.1):**  $\mathbb{E}(ag_1(X) + bg_2(X) + c) = a\mathbb{E}g_1(X) + b\mathbb{E}g_2(X) + c$ .

## Caution

$\mathbb{E}g(X) \neq g(\mathbb{E}X)$  in general. The Cauchy distribution has no mean ( $\mathbb{E}|X| = \infty$ ).

## Jensen

If  $g$  is convex:  $\mathbb{E}g(X) \geq g(\mathbb{E}X)$ . (Concave: reverse.) Equality iff  $g$  is linear on the support of  $X$  or  $X$  is constant.

## Uses

$\mathbb{E}X^2 \geq (\mathbb{E}X)^2$  (so  $\text{Var } X \geq 0$ ).

$X > 0$ :  $\mathbb{E}\frac{1}{X} \geq \frac{1}{\mathbb{E}X}$  and  $\mathbb{E}\ln X \leq \ln \mathbb{E}X$ .

# Moments and variance

## Definitions

$n$ th moment  $\mu'_n = \mathbb{E}X^n$ ;  $n$ th central moment  $\mu_n = \mathbb{E}(X - \mu)^n$ , with  $\mu = \mathbb{E}X$ .

$\text{Var } X = \mathbb{E}(X - \mu)^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2$ ;  $\text{SD} = \sqrt{\text{Var } X}$ .

## Thm 2.3.4

For constants  $a, b$ :  $\text{Var}(aX + b) = a^2 \text{Var } X$ .

The shift  $b$  does not matter; the scale  $a$  enters *squared*.

**Worked example:**  $f(x) = 3x^2$  on  $(0, 1)$

$$\mathbb{E}X = \int_0^1 3x^3 dx = \frac{3}{4}, \quad \mathbb{E}X^2 = \int_0^1 3x^4 dx = \frac{3}{5}, \quad \text{Var } X = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}.$$

$$\mathbb{E}(2X + 1) = 2 \cdot \frac{3}{4} + 1 = \frac{5}{2}, \quad \text{Var}(2X + 1) = 4 \cdot \frac{3}{80} = \frac{3}{20}.$$

# Moment generating functions

## Key facts

$M_X(t) = \mathbb{E}e^{tX}$ , required finite for  $t$  in an open interval around 0.

**Moments (Thm 2.3.7):**  $\mathbb{E}X^n = M_X^{(n)}(0)$ . **Linear change (Thm 2.3.15):**  $M_{aX+b}(t) = e^{bt}M_X(at)$ .

**Uniqueness (Thm 2.3.11b):** if  $M_X(t) = M_Y(t)$  for all  $t$  near 0, then  $X$  and  $Y$  have the same distribution.

**Independent sums:**  $M_{X+Y} = M_X M_Y$  (used to identify  $\sum X_i$ ).

| Distribution              | mgf                            | Domain        |
|---------------------------|--------------------------------|---------------|
| Binomial( $n, p$ )        | $(pe^t + 1 - p)^n$             | all $t$       |
| Poisson( $\lambda$ )      | $e^{\lambda(e^t - 1)}$         | all $t$       |
| Gamma( $\alpha, \beta$ )  | $(1 - \beta t)^{-\alpha}$      | $t < 1/\beta$ |
| Normal( $\mu, \sigma^2$ ) | $e^{\mu t + \sigma^2 t^2 / 2}$ | all $t$       |